

3. A food colouring contains a mixture of yellow and green dyes.  
The table shows the  $R_f$  values of these dyes using different solvents in separate chromatography experiments.

| Colour | $R_f$ value when the solvent is |      |         |
|--------|---------------------------------|------|---------|
|        | Water                           | Acid | Alcohol |
| yellow | 0.74                            | 0.35 | 0.00    |
| green  | 0.76                            | 0.81 | 0.46    |

(a) Why would yellow and green dyes be difficult to separate in a chromatography experiment using **water** as the solvent? [1]

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(b) Describe what you would expect to see in the chromatogram obtained using **alcohol** as the solvent. [1]

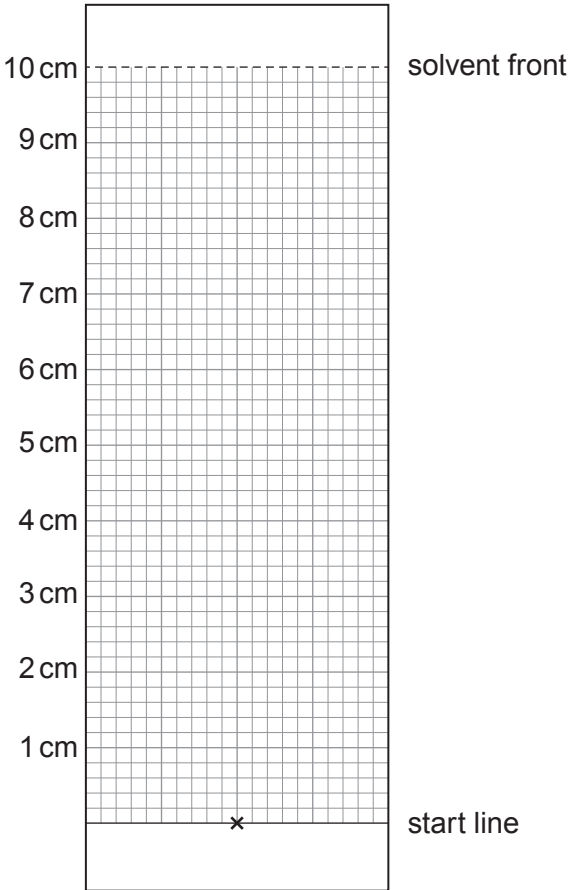
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(c) Complete the chromatogram to show the results you would expect using **acid** as the solvent. [2]



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